The Left-Two Lemma and the Even-Block Gap in the Rank Isolation Theorem

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Abstract

We prove the *Left-Two Lemma*: for any representation of $\mathfrak{S}_n$ and any $k \geq 4$,

$$E_{k-3}^{(+1)} \cap E_{k-1}^{(+1)} \cap \ker(P_{k-2}P_k) = \{0\},$$

where $E_i^{(+1)}$ denotes the $+1$-eigenspace of s_i and $P_i = (1+s_i)/2$. This generalises the double adjacent disjointness argument that closes the even-block gap at $k = 4$ in the Rank Isolation Theorem, and reduces the remaining gap (even blocks $k \geq 6$) to showing that the cascade image at the gap step lies in $E_{k-3}^{(+1)}$. We verify computationally that the cascade image avoids the dangerous subspace for all irreps of $\mathfrak{S}_n$ with $n \leq 9$, and identify the precise obstruction: the intermediate projection P_{k-4} within block $k-2$ breaks $E_{k-3}^{(+1)}$.

1 Setup and context

We work in the setting of the Rank Isolation Theorem (see companion documents `2026-04-15-rank-isolation.te` and `2026-04-16-frobenius-injectivity.tex`).

Definition 1.1. For a representation V of $\mathfrak{S}_n$ and a simple transposition $s_i = (i,\, i{+}1)$, write

$$\begin{aligned} E_i^{(+1)} &= \{v \in V : s_i v = v\}, \\ E_i^{(-1)} &= \{v \in V : s_i v = -v\}, \\ P_i &= \tfrac{1}{2}(1+s_i)\colon V \to E_i^{(+1)}. \end{aligned}$$

Lemma 1.2 (Adjacent disjointness)**.** *For any representation of $\mathfrak{S}_n$ and adjacent indices $|i-j| = 1$:*

$$E_i^{(+1)} \cap E_j^{(-1)} = \{0\} \qquad \textit{and} \qquad E_i^{(-1)} \cap E_j^{(+1)} = \{0\}.$$

Proof. The subgroup $\langle s_i, s_j \rangle \cong \mathfrak{S}_3$ satisfies $(s_i s_j)^3 = 1$. If $s_i v = v$ and $s_j v = -v$, then $(s_i s_j)^3 v = (s_i s_j)^3 v = (-1)^3 v = -v$, contradicting $(s_i s_j)^3 = 1$. □

The staircase product Π for $w_0 \in \mathfrak{S}_n$ processes blocks $B_1, B_2, \ldots, B_{n-1}$ where block k applies $P_k, P_{k-1}, \ldots, P_1$ in sequence. The *Rank Isolation* approach proves $\operatorname{rank}(\Pi|_V) = \dim(V^H)$ (with $H = \langle s_1, s_3, s_5, \ldots \rangle$) by showing:

(a) **Within-block cascade** (proved): each step P_{j-1} after P_j in a block is injective, by adjacent disjointness.

(b) **Odd block starts** (proved): at P_k with k odd ($s_k \in H$), the contraction argument shows the rank drop is exactly the expected one.

(c) **Even block starts**: at P_k with k even ($s_k \notin H$), P_k should be injective on the cascade image. **Proved for $k = 2, 4$; gap for $k \geq 6$.**

The even-block injectivity at k reduces (via cascade transport) to a single step: whether P_{k-2} within block $k-1$'s cascade creates pure $E_k^{(-1)}$ vectors from the image in $E_{k-1}^{(+1)}$.

2 The Left-Two Lemma

Definition 2.1 (Dangerous subspace)**.** For even $k \geq 4$, the *dangerous subspace* is

$$D_k = E_{k-1}^{(+1)} \cap \ker\big(P_{k-2}P_k\big) = \big\{w \in E_{k-1}^{(+1)} : (1 + s_{k-2})(1 + s_k)w = 0\big\}.$$

A vector $w \in E_{k-1}^{(+1)}$ lies in D_k if and only if P_{k-2} maps w into $E_k^{(-1)}$.

Lemma 2.2 (Characterisation of D_k)**.** *If $w \in D_k$, then the $E_k^{(+1)}$ component $w_+ = (w + s_k w)/2$ satisfies $w_+ \in E_{k-2}^{(-1)}$.*

Proof. Since s_{k-2} and s_k commute ($|k - (k-2)| = 2$), the projection P_{k-2} preserves each s_k-eigenspace. Decompose $w = w_+ + w_-$ with $s_k w_\pm = \pm w_\pm$. Then

$$(1 + s_k)w = 2w_+,$$

so $(1 + s_{k-2})(1 + s_k)w = 2(1 + s_{k-2})w_+ = 0$ gives $w_+ \in \ker(1 + s_{k-2}) = E_{k-2}^{(-1)}$. □

Theorem 2.3 (Left-Two Lemma)**.** *For any representation of $\mathfrak{S}_n$ and any $k \geq 4$:*

$$E_{k-3}^{(+1)} \cap E_{k-1}^{(+1)} \cap \ker\big(P_{k-2}P_k\big) = \{0\}.$$

Proof. Let w lie in the triple intersection. By Lemma **??**, $w_+ \in E_{k-2}^{(-1)}$.

Since s_{k-3} commutes with s_k ($|k-3-k| = 3 \geq 2$), the operator s_{k-3} preserves each s_k-eigenspace. In particular, $w \in E_{k-3}^{(+1)}$ implies $w_+ \in E_{k-3}^{(+1)}$.

Now $w_+ \in E_{k-3}^{(+1)} \cap E_{k-2}^{(-1)}$. Since $|(k-3) - (k-2)| = 1$, Lemma **??** gives $w_+ = 0$.

Hence $w = w_- \in E_{k-1}^{(+1)} \cap E_k^{(-1)}$. Since $|(k-1) - k| = 1$, Lemma **??** gives $w = 0$. □

Remark 2.4 (Relation to $k = 4$ proof)*.* For $k = 4$, the theorem reads $E_1^{(+1)} \cap E_3^{(+1)} \cap \ker(P_2P_4) = \{0\}$. At the gap step for block 4, the cascade image is in $E_3^{(+1)}$ (from P_3) and in $E_1^{(+1)}$ (from V^H transport, since $s_1 \in H$ and s_1 commutes with s_3, s_4). So the Left-Two Lemma directly implies P_2 does not create $E_4^{(-1)}$ vectors, closing the $k = 4$ gap. This is exactly the "double adjacent disjointness" argument of the April 15 session.

3 Application and obstruction for $k \geq 6$

3.1 The general pattern

For even $k \geq 6$, the Left-Two Lemma would close the gap if the cascade image at the gap step were in $E_{k-3}^{(+1)}$. The cascade image at the gap step is $I = P_{k-1}(J)$, where J is the output of blocks $1, \ldots, k-2$.

Proposition 3.1 (Eigenspace tracking)**.** *The cascade image I at the gap step satisfies:*

(i) $I \subset E_{k-1}^{(+1)}$ *(from the just-applied P_{k-1}).*

(ii) $I \subset E_1^{(+1)}$ *(since $s_1 \in H$, and s_1 commutes with all generators $s_2, \ldots, s_{k-1}$ used in the cascade for $k \geq 5$).*

(iii) $I \not\subset E_{k-3}^{(+1)}$ *in general.*

Proof. (i) and (ii) follow from the cascade structure: every block ends with P_1, so $J \subset E_1^{(+1)}$; and s_1 commutes with s_{k-1} (distance ≥ 3 for $k \geq 5$), so P_{k-1} preserves $E_1^{(+1)}$.

For (iii): within block $k-2$, the cascade applies $P_{k-2}, P_{k-3}, P_{k-4}, \ldots, P_1$. After P_{k-3}: the image is in $E_{k-3}^{(+1)}$. But P_{k-4} uses s_{k-4}, which is adjacent to s_{k-3} ($|k-4-(k-3)| = 1$), so P_{k-4} does not preserve $E_{k-3}^{(+1)}$. The subsequent steps $P_{k-5}, \ldots, P_1$ and P_{k-1} all commute with s_{k-3} and therefore preserve the (broken) s_{k-3} eigenspace status. So $E_{k-3}^{(+1)}$ is not recovered. □

3.2 Computational evidence

Proposition 3.2 (Verified avoidance)**.** *For all irreducible representations V_λ of $\mathfrak{S}_n$ with $n \leq 9$ and all even k with $6 \leq k \leq n-1$:*

$$I \cap D_k = \{0\}$$

where I is the cascade image and D_k the dangerous subspace. In particular, P_k is injective on the cascade image at every even block start.

Combined with parabolic reduction (each even k reduces to $\mathfrak{S}_{k+1}$ irreps) and earlier computational verification through $k = 14$, the Rank Isolation Theorem holds unconditionally for $n \leq 16$.

3.3 Nature of the obstruction

The avoidance $I \cap D_k = \{0\}$ holds despite the cascade image *not* lying in $E_{k-3}^{(+1)}$ and *not* having its $E_k^{(+1)}$ components in $E_{k-2}^{(+1)}$. Specifically:

- The $E_{k-3}^{(-1)}$ component of vectors in I can be substantial (ratios $\|v^{(-)}\|/\|v\|$ up to 0.87 at $k = 8$).
- The $E_{k-2}^{(-1)}$ component of $w_+ = (v + s_k v)/2$ is generically nonzero.
- The avoidance is *not* explained by orthogonality (I is not in $D_k^\perp$ within $E_{k-1}^{(+1)}$).

The mechanism appears to involve the *global* structure of the cascade image — the full history of projections constrains the image to a subspace that is transverse to D_k for reasons not captured by any single eigenspace condition.

4 Summary of proof architecture

Component	**Status**	**Method**
Within-block cascade	Proved	Adjacent disjointness
Block 2 start (P_2)	Proved	Adjacent disjointness
Odd block starts (P_k, k odd)	Proved	Contraction argument
Even block $k = 4$ (P_4)	Proved	Left-Two Lemma with $E_1^{(+1)}$
Even blocks $k \geq 6$	**Gap**	Left-Two Lemma + $E_{k-3}^{(+1)}$?
Unconditional range	$n \leq 16$	Parabolic reduction + computation

The Left-Two Lemma reduces the even-block gap to a single question:

Conjecture 4.1 (Cascade $E_{k-3}^{(+1)}$ preservation)**.** *There exists a proof that the cascade image at the even-block gap step avoids D_k for all k, either by:*

- *finding a condition weaker than $E_{k-3}^{(+1)}$ that the cascade image satisfies and that is sufficient to exclude D_k;*
- *an inductive argument using the proved $k = 4$ case as base;*
- *a global argument (e.g., total rank in the regular representation) that bypasses block-by-block analysis.*

5 Verification

The Left-Two Lemma was verified computationally for all irreps of $\mathfrak{S}_n$ with $n = 7, 8, 9$. The "left-two" intersection $E_{k-3}^{(+1)} \cap E_{k-1}^{(+1)} \cap \ker(P_{k-2}P_k)$ is $\{0\}$ in every case (75+ cases tested), confirming the algebraic proof. The "standard" intersection $E_1^{(+1)} \cap E_{k-1}^{(+1)} \cap \ker(P_{k-2}P_k)$ is *not* $\{0\}$ for most irreps, confirming that $E_1^{(+1)}$ is insufficient.

Scripts: `even_block_gap_analysis.sage`, `cascade_eigenspace_detail.sage`, `even_block_reverse_cascade.`